Replication Report: Exactly-Solved Model of Light-Scattering Errors in Quantum Simulations with Metastable Trapped-Ion Qubits

OSTI 2441075 | Coverage 8/10, Agreement 9/10

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2026-04-27

1 Source paper

- **Title:** Exactly-Solved Model of Light-Scattering Errors in Quantum Simulations with Metastable Trapped-Ion Qubits
- **Authors:** Foss-Feig, Vikas, et al.
- **Year:** 2024
- **Domain:** Quantum Information / AMO
- **Identifier:** OSTI 2441075

2 Replication objective

The central claim of the paper concerns quantum information / amo. Our objective was to reproduce the headline numerical/qualitative results within the constraints of the AI-driven Replication Project (24–72 h, commodity GPU/CPU compute, no author contact). A replication plan is archived at `2441075-Exactly-solved-model-of-light-scattering-errors-in/replication_plan.pdf`.

3 Methodology

We followed the standard Replication Project workflow: ingest the source PDF, extract method and data pointers, draft a replication plan, attempt the smallest end-to-end pipeline that produces a comparable number, then scale up where compute and data permit. Tools used in this replication are catalogued in the meta-paper's computational tools inventory (see `COMPUTATIONAL_TOOLS_INVENTORY.md`).

Paper-directory README excerpt

Exactly-solved model of light-scattering errors in quantum simulations with metastable trapped-ion qubits

- **OSTI ID:** 2441075
- **Rank:** #24
- **Replication Score:** 9/10

- **Open-Source Tools:** Yes
- **Code Repository:** No
- **Tools:** Python (implied for numerical integration), Simpson's method (for ODEs)

Why This Paper

Analytic/numerical, all equations/algorithms specified, synthetic data, and quantitative validation. No code repo but trivial for AI.

Replication Plan

AI extracts formulas/algorithms, implements Lindbladian master equation, generates synthetic data, and reproduces results.

Status

- [ ] Paper reviewed
- [ ] Code/tools identified
- [ ] Code implemented/cloned
- [ ] Results reproduced
- [ ] Results validated against paper

The replication was driven by an OpenClaw agent loop (Claude Opus / GPT-5 class models) issuing shell, Python, and (where applicable) GPU jobs. Compute usage and wall-time for individual papers are summarised in the meta-paper's resource table; not separately recorded for this paper unless noted in the Tier-Lift artefact. Sub-agents were spawned for replication-plan generation and (for papers in batches 1–4) for tier-lift extension runs.

4 What was reproduced

- Full analytic solution Eqs. 6-10 with all four I, R, L, B components (including the paper's novel L and B terms)
- 5-jump-operator Lindblad numerical master equation with correct factor-of-2 jump convention
- Ca branching ratios (94.5% leakage / 3.9% Raman / 1.6% elastic)
- Selection rule: |1 =|mJ=-5/2 dark under polarization
- Machine-precision analytic vs numerical agreement (10 ^{11}–10 ^{13})
- Figure 2 GHZ-fidelity qualitative trends (N dependence, postselection gain, B-field effect)
- Figure 3 correlation decay (faster than exp(-mΓt)) and perpendicular-correlation growth
- Figure 4 spin-squeezing ^{2} N^(-2/3) with scattering floor

5 What was missing or incomplete

- Penning-trap 2D triangular-lattice ion-crystal mode structure
- Full 3D normal-mode J__ij computation (we used 1D chain)
- Laser power P and beam waist w (not specified in paper)
- Exact trap frequencies z, r (referenced to Ref [20])
- GHZ fidelity fourth-order J__ij correction terms
- Quantitative N-thresholds from Fig 2 match

- Experimental validation against real Ba or Ca data

The dominant blocker categories for this paper, mapped onto the meta-paper's 9-category friction taxonomy (§4), are listed in §?? below.

6 Quantitative agreement

Per-quantity numerical comparisons are not separately tabulated in the structured evaluation record for this paper beyond the agreement rationale below; where the rationale references specific numbers, they are reproduced verbatim. A full numerical comparison table is recorded in the per-replication artefacts (see §??). For papers below 8/8, the agreement rationale identifies the specific quantities that diverge.

Agreement rationale. Analytic-numerical agreement at machine precision (errors 10^{11}–10^{13} across all tested cases). Trace preservation to 5.5×10^{1}, positivity to 10^{1} — essentially exact. All qualitative figure features correct: fidelity ↓ with N, Postselection helps, m-body correlations decay faster than exp(-mΓt), ^{2} N^(-2/3) scaling with scattering floor. Quantitative figure values differ because paper's Penning-trap J_ij differs from our 1D-chain J_ij.

7 Score and friction-point tagging

- **Coverage:** 8/10 — Complete implementation of analytic solution (Eqs. 6-10) with all four terms I, R, L, B — including the paper's novel leakage (L) and combined (B) contributions. Five-jump-operator Lindblad master equation implemented numerically with correct factor-of-2 jump convention. Ca branching ratios (94.5% leakage / 3.9% Raman / 1.6% elastic). Figures 1 (schematic), 2 (GHZ fidelity), 3 (correlations), 4 (squeezing) reproduced qualitatively. Missing: full Penning-trap 2D crystal mode computation (we used 1D chain approximation for J_ij).
- **Agreement:** 9/10 — Analytic-numerical agreement at machine precision (errors 10^{11}–10^{13} across all tested cases). Trace preservation to 5.5×10^{1}, positivity to 10^{1} — essentially exact. All qualitative figure features correct: fidelity ↓ with N, Postselection helps, m-body correlations decay faster than exp(-mΓt), ^{2} N^(-2/3) scaling with scattering floor. Quantitative figure values differ because paper's Penning-trap J_ij differs from our 1D-chain J_ij.

Friction points encountered (taxonomy tags):

- **F7**: Missing surrogate calculators (xTB/DFT/closed solver)

8 Follow-on questions

- Q1: Implement the full 2D Penning-trap equilibrium-crystal mode solver and recompute J_ij — do the Fig 2 fidelity crossovers match the paper's reported N_threshold at B=0.9T and B=4.5T within 10%?
- Q2: Extend the exact solution from to $\pm$ polarization where |1 is no longer dark — derive new analytic expressions and quantify the fidelity penalty of losing the dark-state protection.
- Q3: Test the exact solution against a full Lindblad simulation at N=20 (near the boundary of tractable numerics) — does the product-form correlation factorization still agree to machine precision, or do many-body corrections emerge?

- Q4: For ^{13}Ba (different branching ratios, ~70% leakage), do the novel L and B terms still dominate the error budget, and does postselection recover the same 5.5% residual-error advantage?
- Q5: Combine the exact scattering model with realistic single-qubit gate errors (miscalibration, Stark shifts) to produce a full end-to-end error budget for a 50-qubit GHZ preparation circuit — at what N does gate error overtake scattering as the dominant error source?

9 Reproducibility pointers

- **Paper directory:** `2441075-Exactly-solved-model-of-light-scattering-errors-in`
- **Replication artefacts:**
  - `/Users/stevens/Dropbox/REPLICATE-PROJECT/2441075-Exactly-solved-model-of-light-scatte`
- **Replication-dir hint from JSONL:** `~/Dropbox/REPLICATE-PROJECT/2441075-Exactly-solved-model-`
- **Status:** Not recorded

To re-run, change directory into the paper directory above and consult its `README.md` for the entry-point script. Most replications follow the convention `replication/run_*.sh` or `replication/<subdir>/run.py`.